# AAE 54800: Mechanical Behavior of Aerospace Materials

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| **Credit Hours** | 3 |
| **Offered** | Fall |
| **Pre-requisites** | AAE 204 or equivalent |
| **Co-requisites** | None |
| **Instructional Method** | 3 hours of lecture per week |
| **Required** | No |

## 1. Course Description

This course serves as an overview for materials behavior for students without a materials background, including seniors and entry-level graduate students. Materials are at the foundation for all of engineering, as evident by the latest products that we design, to the airplanes that we fly, to the latest smart phones. In fact, breakthroughs with material research are often accompanied by rapid advancements in technology. Thus, it is paramount for all engineers to have an understanding of the structure and behavior of materials.

## 2. Instructor Information

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## 3. Topics Covered

- Structure of materials, the microstructure connection to mechanical properties, and ultimately failure mechanisms. Materials play an important role in both design and manufacturing, which will be addressed in the context of components and extreme environments.
- Defects within materials, defect formation/evolution, and their role in strengthening mechanisms.
- Material anisotropy, micromechanisms, and elasto-plastic properties at the atomic, single crystal/constituent, and polycrystal/material levels and their use in explaining the deformation and failure characteristics in metals, polymers, and ceramics;
- Failure mechanisms and toughening in composites; structure and behavior of aerospace materials: metal alloys, ceramic-matrix composites, and fiber-reinforced polymer composites.
- Elastic deformation, dislocation mechanics, plastic deformation and strengthening mechanisms, creep, and failure mechanisms; design criteria; special topics.

## 4. Intended Learning Outcomes

On completing this course, the student shall be able to:

- Ability to express vector and tensor equations using indicial notation.
- Ability to express crystallography according to Miller indices and define slip systems in basic crystal structures.
- Ability to calculate and understand the physical basis of elastic anisotropy.
- Ability to understand, calculate, and use basic states of stress and yield criteria.
- Ability to define and understand the physical basis of point, line, and area defects in crystalline materials.
- Ability to calculate the strain fields, stress fields, and energy of dislocations, in addition forces between dislocations.
- Ability to understand dislocation motion as well as interactions between dislocations.

- Ability to resolve shear stress on slip systems during deformation of single crystals and calculate the velocity gradient.
- Ability to apply equilibrium and compatibility constraints to plastic flow of polycrystalline materials.
- Ability to understand origins of twinning, stacking faults, and the shape memory effects.
- Ability to understand physics and calculate strengthening influence of solutes, precipitates, grain boundaries, and strain hardening.
- Ability to understand physical origins of creep, calculate creep using Larson-Miller expressions, and use creep deformation mechanism map.
- Ability to understand origins of residual stress and apply to constrain problems.
- Ability to understand and apply statistical approaches to identify probability of failure and property variations.
- Ability to define and classify polymer structures.
- Ability to understand viscoelasticity and calculate behavior using classical models.
- Ability to distinguish physical mechanisms of polymer deformation, crazing, and fracture.
- Ability to express stiffness and strength of fiber reinforced composite structures.

## 5. Assessment Method

Weekly homework, midterm exam and final exam.

## 6. Relation to ABET Outcomes

| | Program Learning Outcomes | Included? |
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| **1.** | An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics. | Yes |
| **2.** | An ability to apply engineering design to produce solutions that meet specified needs with consideration | Yes |

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| | of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. | |
| **3.** | An ability to communicate effectively with a range of audiences. | No |
| **4.** | An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts. | Yes |
| **5.** | An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives. | Yes |
| **6.** | An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions. | Yes |
| **7.** | An ability to acquire and apply new knowledge as needed, using appropriate learning strategies. | Yes |